

DAEN 427 Homework 2

Due date: Aug 26, 2025

1. From the following expressions, which one corresponds to the sentence: *The probability of being sunny given that it is June 14th of 1976?*
 - (a) $p(\text{sunny})$
 - (b) $p(\text{sunny} \mid \text{June})$
 - (c) $p(\text{sunny} \mid \text{June 14th of 1976})$
 - (d) $p(\text{June 14th of 1976} \mid \text{sunny})$
 - (e) $p(\text{sunny, June 14th of 1976})/p(\text{June 14th of 1976})$

Answer:

2. Show that the probability of choosing a human at random and picking the Pope is not the same as the probability of the Pope being human. In the animated series Futurama, the (Space) Pope is a reptile. How does this change your previous calculations?

Answer:

3. Hand sketch what the distribution of possible observed values could be for the following cases:
- (a) The number of people visiting your local cafe assuming Poisson distribution.
 - (b) The weight of adult dogs in kilograms assuming a Uniform distribution.
 - (c) The weight of adult elephants in kilograms assuming Normal distribution.
 - (d) The weight of adult humans in pounds assuming skew Normal distribution.

Answer:

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